

ENDOGENEITY IN COUNT DATA MODELS: AN APPLICATION TO DEMAND FOR HEALTH CARE

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SUMMARY

The generalized method of moments (GMM) estimation technique is discussed for count data models with endogenous regressors. Count data models can be specified with additive or multiplicative errors. It is shown that, in general, a set of instruments is not orthogonal to both error types. Simultaneous equations with a dependent count variable often do not have a reduced form which is a simple function of the instruments. However, a simultaneous model with a count and a binary variable can only be logically consistent when the system is triangular. The GMM estimator is used in the estimation of a model explaining the number of visits to doctors, with as a possible endogenous regressor a self-reported binary health index. Further, a model is estimated, in stages, that includes latent health instead of the binary health index. © 1997 by John Wiley & Sons, Ltd.

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1. INTRODUCTION

In many economic applications the variable of interest is a count process, for example the number of visits to the doctor, job applications, or patent applications. The modelling and estimation of such inherently nonlinear processes are well established nowadays, with early references Gourieroux, Monfort, and Trognon (1984), Cameron and Trivedi (1986), and McCullagh and Nelder (1983, 1989). Recent developments are surveyed by Gurmu and Trivedi (1994) and Winkelmann and Zimmermann (1995).

In microeconomic applications, explanatory variables are often simultaneously determined with the dependent variable, resulting in coefficient estimates that are inconsistent when obtained by standard methods. Techniques for dealing with simultaneity in count data regression models are largely underdeveloped when compared to the continuous data case. In a recent paper, Mullahy (1996) discusses instrumental variables, or generalized method of moments, and two-stage estimation methods for Poisson regression models in a specification where unobserved heterogeneity is correlated with the regressors. Testing for exogeneity has been discussed earlier by Grogger (1990), who proposed use of the Hausman test after estimation of the model by nonlinear instrumental variables. Terza (1996) and Neslusan and Terza (1996) derive consistent estimators in nonlinear regression models with endogenous binary variables under parametric assumptions of the error processes.

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In this paper we discuss the generalized method of moments (GMM) estimator for count data models with endogenous regressors, utilizing first-order moment conditions only. Given a set of exogenous instruments for the endogenous variables, the GMM estimation method provides consistent estimates for the parameters.

Count data regression models can be specified with additive or multiplicative errors. These specifications are, in principle, observationally equivalent when only the first-order conditional moment is specified. Differences arise, however, when the choice of instruments in the two specifications under endogeneity is considered, as the same set of instruments is, in general, not orthogonal to both types of errors.

For nonlinear models, the reduced form for endogenous regressors often has a very complicated structure. This is also the case for the reduced form of simultaneous equation models that include a dependent count variable. A specification we analyse and use for the modelling of demand for health care is that of a binary variable which is simultaneously determined with the count variable. Such a system is coherent in the sense of Blundell and Smith (1994) only when it is triangular. For this model, an alternative to GMM estimation, which amounts to instrumenting the binary variable, is the estimation in two stages of a model which is specified in terms of a latent continuous variable. This latent variable determines the binary outcome by means of a threshold transformation.

The model specifications and estimators are applied to a model for visits to or by a doctor (general practitioner) in the last month before the interview, utilizing data from the British Health and Lifestyle Survey 1991–1992. Models that explain the number of visits to the doctor have also been analysed by Cameron *et al.* (1988) and Pohlmeier and Ulrich (1995) for Australian and German data respectively. Factors such as income and education are likely to have a direct effect on demand for medical care, but are also important determinants of health, which in turn affects demand. In order to estimate the direct effects of income and education a measure of health has to be included in the model. The measure we consider is a self-reported health index that is likely to have measurement error which is correlated with the number of visits to the doctor. We therefore instrument the binary health index by estimated reduced-form probabilities. Finally, a model is estimated which includes underlying latent health instead of the binary health index.

The outline of the paper is as follows. In Section 2 the GMM estimation technique is presented. Section 3 discusses simultaneous models and Section 4 presents the estimation results for the demand for doctors. Section 5 presents conclusions.

2. GENERALIZED METHOD OF MOMENTS ESTIMATION

Before discussing the estimation of count data models with endogenous regressors by generalized method of moments we first present the standard Poisson model for count data.

Let $y_i, i = 1, \dots, N$, denote the dependent count variable, which is independently Poisson distributed, with the conditional mean specified as

$$E(y_i | x_i) = \mu_i = \exp(x_i' \beta) \quad (1)$$

where x_i is a k -vector of explanatory variables and β is a k -vector of parameters. The maximum likelihood estimator, denoted $\hat{\beta}_{ML}$, solves the first-order condition $X'(y - \mu) = 0$, where X is an $N \times k$ matrix, and y and μ are N -vectors, and $\sqrt{N}(\hat{\beta}_{ML} - \beta)$ has a limiting normal distribution with mean zero and variance the limit of $(N^{-1}X'MX)^{-1}$, where $M = \text{diag}(\mu_i)$. In practice, the

standard errors are often biased due to the presence of over- or underdispersion. Correct standard errors in these cases are computed from the estimated variance of the Poisson pseudo-likelihood estimator $\hat{\beta}_{\text{PL}} (= \hat{\beta}_{\text{ML}})$ (Gourieroux, Monfort, and Trognon, 1984):

$$\widehat{\text{Var}}(\hat{\beta}_{\text{PL}}) = (X' \hat{M} X)^{-1} \left(\sum_i (y_i - \hat{\mu}_i)^2 x_i x_i' \right) (X' \hat{M} X)^{-1}$$

where $\hat{M} = \text{diag}(\hat{\mu}_i)$ and $\hat{\mu}_i = \exp(x_i' \hat{\beta}_{\text{PL}}) = \exp(x_i' \hat{\beta}_{\text{ML}})$.

The conditional mean specification (1) implicitly defines a regression model

$$y_i = \mu_i + u_i = \exp(x_i' \beta) + u_i$$

with $E(u_i | x_i) = 0$. The generalized method of moments (GMM) estimator (see Hansen, 1982; Ogaki, 1993), based on this moment condition only, minimizes

$$(y - \mu)' X W_N^{-1} X' (y - \mu) \quad (2)$$

where W_N is a weight matrix. As the minimum of equation (2) is obtained at $X'(y - \mu) = 0$, the GMM estimator for β will be the same as the Poisson maximum likelihood estimator, irrespective of the weight matrix. Let $\text{Var}(X'(y - \mu)) = X' \Omega X$. When the variance equals the mean, Ω is equal to M and the variance of the GMM estimator is equal to the variance of the Poisson maximum likelihood estimator. The variance of the GMM estimator is equivalent to the pseudo-likelihood variance when $X' \Omega X$ is estimated by $\sum_i (y_i - \hat{\mu}_i)^2 x_i x_i'$.

When some elements of x_i are endogenous, implying that

$$E(u_i | x_i) \neq 0$$

μ_i is no longer the conditional mean of y_i and the Poisson ML estimator will be inconsistent. If there are instruments z_i available such that

$$E(u_i | z_i) = 0$$

then the consistent nonlinear instrumental variables (NLIV) estimator is given by minimization of (see Amemiya, 1985)

$$(y - \mu)' Z (Z' Z)^{-1} Z' (y - \mu)$$

and is a one-step GMM estimator.¹ The efficient two-step GMM estimator, given the instruments, is found by minimization of

$$(y - \mu)' Z (\widetilde{Z' \Omega Z})^{-1} Z' (y - \mu)$$

where

$$\widetilde{Z' \Omega Z} = \sum_{i=1}^N (y_i - \tilde{\mu}_i)^2 z_i z_i'$$

¹ The NLIV estimator clearly does not take into account the heteroscedasticity of u . A 'Poisson' type first-round estimator is obtained by minimizing $(y - \mu)' Z (Z' M Z)^{-1} Z' (y - \mu)$, which should be iterated until convergence.

is an estimate of the (asymptotic) variance of $Z'(y - \mu)$, with $\tilde{\mu}_i = \exp(x_i'\tilde{\beta})$, and $\tilde{\beta}$ is the first-round estimate of β . Denote the two-step GMM estimator by $\hat{\beta}_{\text{GMM2}}$. Under standard regularity assumptions, the limiting distribution of $\sqrt{N}(\hat{\beta}_{\text{GMM2}} - \beta)$ is normal with mean zero and variance the limit of

$$(N^{-1}(X'MZ)(Z'\Omega Z)^{-1}(Z'MX))^{-1}$$

Optimal instruments, which minimize the variance of the two-step GMM estimator, are given by (see, for example, Davidson and MacKinnon, 1993)

$$Z^* = E(\Omega^{-1}D|Z)$$

where D is the matrix of derivatives $\partial(y - \mu)/\partial\beta$, which is equal to $-MX$. The optimal instruments therefore are

$$Z^* = E(\Omega^{-1}MX|Z)$$

When $Z = X$, i.e. there is no endogeneity, the optimal instruments for the Poisson model are given by X . For $Z \neq X$, it is in general impossible to get consistent estimates of Z^* . A reasonable working hypothesis may be to specify $\Omega = M$, which leads to the use of the instruments $\hat{X} = E(X|Z)$. A test for overidentifying restrictions can be obtained by augmenting the instruments $\hat{X}$ by the elements of Z different from and not collinear with $\hat{X}$, and using the standard test for overidentifying restrictions for the GMM estimator that utilizes the augmented instrument matrix.

2.1. Additive versus Multiplicative Errors

A multiplicative model is specified as

$$y_i = \exp(x_i'\beta + \tau_i) = \exp(x_i'\beta)v_i = \mu_i v_i, \quad v_i = \exp(\tau_i) \quad (3)$$

which is motivated by treating the unobservables τ_i and observables x_i symmetrically. In principle, the multiplicative and additive models are observationally equivalent when only the first-order conditional moment is specified (see Wooldridge, 1992). Differences arise, however, when the choice of instruments in the two specifications under endogeneity is considered, as the same set of instruments is, in general, not orthogonal to both type of errors.

Endogeneity implies that in equation (3) $E(v_i|x_i) \neq 1$. Let z_i be a set of instruments that satisfies $E(v_i|z_i) = 1$, then an instrumental variable estimator can be based on the residual $v_i - 1$, which is equal to $(y_i - \mu_i)/\mu_i = u_i/\mu_i$. Under endogeneity the same set of instruments cannot, in general, be orthogonal to u_i and u_i/μ_i at the same time, i.e. when $E(u_i|z_i) = 0$, it follows in general that $E(u_i/\mu_i|z_i) \neq 0$ due to the correlation between u_i and μ_i .

Which transformation should be used in the GMM estimation when there is endogeneity present is an empirical issue which can be tested by the standard test for overidentifying restrictions in cases where there are more instruments than endogenous variables. The two-step GMM estimator in the multiplicative model minimizes the objective function

$$(y - \mu)'M^{-1}Z(\widetilde{Z'\Omega^*Z})^{-1}Z'M^{-1}(y - \mu)$$

where $Z'\Omega^*Z$ is the asymptotic variance of $Z'M^{-1}(y - \mu)$. When $Z = X$ in the Poisson model, with $\Omega^* = M^{-1}$, this becomes

$$(y - \mu)'M^{-1}X(X'M^{-1}X)^{-1}X'M^{-1}(y - \mu) \quad (4)$$

which is equivalent to a heteroscedasticity corrected objective function. Clearly, equation (4) will not yield Poisson ML estimates. However, the optimal instruments for the multiplicative model are given by

$$Z^* = E(\Omega^{*-1}WX | Z)$$

where $W = \text{diag}(y_i/\mu_i)$. When $Z = X$ and $\Omega^* = M^{-1}$, this reduces to $Z^* = MX$ and the estimator minimizing equation (4) with MX as instruments is the same as the Poisson ML estimator. Equivalently, if Z are valid instruments in the additive model, then MZ are valid instruments in the multiplicative model, giving the same estimation results for the two model specifications.

Mullahy (1996) proposed use of the transformed residual u_i/μ_i in a model with unobserved heterogeneity which is correlated with (some of) the regressors. His model is specified as

$$y_i = \exp(x_i'\beta)\eta_i + \varepsilon_i = \mu_i\eta_i + \varepsilon_i \quad (5)$$

where η_i is the unobserved heterogeneity term and

$$E(\varepsilon_i | x_i) = 0, \quad E(\eta_i | x_i) \neq 1$$

Clearly, model (5) is observationally equivalent to the multiplicative model (3) with endogeneity.

3. SIMULTANEOUS EQUATIONS

Specifying a simultaneous equations model where one dependent variable is a count, and which has a reduced form with a simple structure, is virtually impossible. Consider, for example, a two-equation model with a count variable and a continuous variable that are simultaneously determined. The system of equations can be specified as

$$y_{1i} = \exp(\alpha y_{2i} + x'_{1i}\beta) + u_{1i}$$

$$y_{2i} = \gamma y_{1i} + x'_{2i}\delta + u_{2i}$$

and the reduced form for y_{2i} has to be derived from

$$y_{2i} = \gamma \exp(\alpha y_{2i} + x'_{1i}\beta) + x'_{2i}\delta + \gamma u_{1i} + u_{2i}$$

which does not reduce to a simple equation for y_{2i} .

A model specification we are particularly interested in, given the application in the next section, is that of a simultaneous model with a count and a binary variable as endogenous regressors. Two possible model specifications are considered, one with the endogenous binary variable and one with an unobserved latent variable as covariates in the model for the count variable. These two models are referred to as Model 1 and Model 2, respectively.

3.1. Model 1: Binary Endogenous Variable

Model 1 is specified as

$$\begin{aligned} y_{1i} &= \exp(\alpha y_{2i} + x'_{1i}\beta) + u_{1i} \\ y_{2i}^* &= \gamma y_{1i} + x'_{2i}\delta + u_{2i} \\ \text{Cov}(u_{1i}, u_{2i}) &= \begin{bmatrix} \sigma_{1i}^2 & \sigma_{12i} \\ \sigma_{12i} & 1 \end{bmatrix} \end{aligned}$$

where y_{2i}^* is an unobserved latent variable. The binary variable y_{2i} is related to y_{2i}^* by

$$\begin{aligned} y_{2i} &= 1 \quad \text{if } y_{2i}^* > 0 \\ y_{2i} &= 0 \quad \text{otherwise} \end{aligned}$$

and σ_2^2 is normalized to 1. Clearly, σ_{1i} and σ_{12i} can be functions of the regressors and parameters.

For this model, the logical consistency, or coherency, is an issue. Following Maddala (1983, p. 118) and Blundell and Smith (1994) it is easily established that the model is only coherent when $\gamma = 0$ or $\alpha = 0$ as

$$\begin{aligned} P(y_{2i} = 1) + P(y_{2i} = 0) &= F(\gamma \exp(\alpha + x'_{1i}\beta) + x'_{2i}\delta) + (1 - F(\gamma \exp(x'_{1i}\beta) + x'_{2i}\delta)) \\ &= 1 \quad \text{if } \gamma = 0 \quad \text{or} \quad \alpha = 0 \end{aligned}$$

So, if $\alpha \neq 0$, i.e. a binary variable is included as regressor in the model for the count variable, we have that for coherency $\gamma = 0$, and endogeneity occurs when $\sigma_{12i} \neq 0$. In the following discussion these conditions are assumed to hold.

Estimation in stages, by replacing y_{2i} by its estimated conditional mean, does not give consistent estimates of the parameters. If the conditional mean of y_{2i} is specified as

$$E(y_{2i} | x_{2i}) = F(x'_{2i}\delta)$$

with F a CDF, then estimation in stages of the equation

$$y_{1i} = \exp(\alpha F(x'_{2i}\delta) + x'_{1i}\beta) \exp(\alpha w_{2i}) + u_{2i}$$

where $w_{2i} = y_{2i} - F(x'_{2i}\delta)$, leads to inconsistent results due to the fact that the moments of w_{2i} are dependent on x_{2i} , as $E(w_{2i}^2 | x_{2i}) = F(x'_{2i}\delta)(1 - F(x'_{2i}\delta))$, and so $E(\exp(w_{2i}))$ is a function of the parameters and regressors.

A consistent estimator for (α, β) in Model 1 is the GMM estimator, and a natural choice of instrument for y_{2i} is $F(x'_{2i}\hat{\delta})$, where $\hat{\delta}$ is the logit or probit estimator of δ .

Equivalently, Model 1 could be specified with a multiplicative error in the count specification as

$$\begin{aligned} y_{1i} &= \exp(\alpha y_{2i} + x'_{1i}\beta + u_{1i}) + \varepsilon_i \\ y_{2i}^* &= x'_{2i}\delta + u_{2i} \end{aligned}$$

Terza (1996) derives consistent estimators for the parameters under the assumption of joint normality of u_1 and u_2 , by adjusting the mean of the count specification for the conditional expectation of $\exp(u_1)$ given that $(u_2 > -x'_2\delta)$ or $(u_2 \leq -x'_2\delta)$. GMM estimation of this multiplicative specification has been discussed in Section 2.1.

3.2. Model 2: Latent Endogenous Variable

Model 2 is specified as

$$\begin{aligned} y_{1i} &= \exp(\alpha y_{2i}^* + x'_{1i}\beta) + u_{1i}^* \\ y_{2i}^* &= \gamma y_{1i} + x'_{2i}\delta + u_{2i} \\ \text{Cov}(u_{1i}^*, u_{2i}) &= \begin{bmatrix} \sigma_{1i}^{*2} & \sigma_{12i}^* \\ \sigma_{12i}^* & 1 \end{bmatrix} \end{aligned} \quad (6)$$

As discussed above, the reduced form of y_{2i}^* is a complicated function of the exogenous variables. The parameters cannot be estimated by GMM as y_{2i}^* is not observed. The model may, however, be estimated by substituting some reduced form of y_{2i}^* into the model for y_{1i} . For example, if $\gamma = 0$, equation (6) can be written as

$$y_{1i} = \exp(\alpha x'_{2i}\delta + x'_{1i}\beta) \exp(\alpha u_{2i}) + u_{1i}^* \quad (7)$$

Consider δ known, then if u_{2i} is statistically independent of x_{2i} and x_{1i} , and $E[u_{1i}^* | x_{1i}, x_{2i}] = 0$, the conditional mean of y_{1i} given x_{1i} and $x'_{2i}\delta$ is given by

$$E[y_{1i} | x_{1i}, x'_{2i}\delta] = \exp(\alpha x'_{2i}\delta + x'_{1i}\beta^*) \quad (8)$$

where β^* is equal to β , but with the constant shifted by $\ln(E[\exp(\alpha u_{2i})])$. So in this case the Poisson pseudo-likelihood estimator is a consistent estimator for α and β^* . As δ is unknown, one way to estimate α is to estimate the model in stages. The first stage is to estimate δ by logit or probit. The second stage is to substitute the estimator for δ into equation (7) and to estimate the model by Poisson pseudo-likelihood. The standard errors of such an estimator have to be adjusted to take account of the estimation in stages (see Appendix).

4. DEMAND FOR HEALTH CARE

In this section we estimate models for the demand for health care in terms of the number of visits to the doctor. A self-reported health index is included as a regressor, which is likely to be endogenous.

The data are taken from the British Health and Lifestyle Survey 1991–1992 (HALS2). This survey is a follow-up from a previous survey in 1984–5. In the first survey the people interviewed were 18 years and older, and so the minimum age of the people interviewed in the second survey is 25. Of the original sample, 59% were interviewed in the follow-up. The drop-out rates between the two samples, due to death, refusal, and non-tracing, affect the sampling distribution, and the HALS2 survey cannot be considered to be a representative sample of the adult population of 25 years and older (see HALS2 User Guide). For example, a higher proportion of interviews was achieved in non-manual than in manual groups. However, the age/sex distribution of the HALS2

Table I. Summary statistics

Var	Description	Mean	St. dev.	Min	Max
Visits	# visits to/by a GP in the month before interview	0.4026	0.8059	0	10
Male	Sex = male	0.4348	0.4958	0	1
Age	Age	51.32	15.87	25	96
Age ²	Age × Age/100	28.86	17.22	6.25	92.16
Single	Single	0.0956	0.2940	0	1
Unem	Unemployed	0.0270	0.1621	0	1
<i>Education</i>					
Edu1	CSE Grades 1–5	0.1664	0.3724	0	1
Edu2	GCE 'A' level	0.0357	0.1856	0	1
Edu3	ONC/OND/HNC/HND	0.1388	0.3457	0	1
Edu4	Teacher/Nurse	0.0538	0.2256	0	1
Edu5	Professional/Degree	0.1248	0.3306	0	1
Edu6	Other	0.0372	0.1893	0	1
<i>After-tax weekly personal income (£)</i>					
Inc1	(.,100)	0.4508	0.4976	0	1
Inc2	[100,250)	0.3637	0.4811	0	1
Inc3	[250,400)	0.1359	0.3427	0	1
Inc4	[400,.)	0.0496	0.2172	0	1
<i>Short-term health</i>					
Pregnant	Pregnant at time of interview	0.0058	0.0761	0	1
Tempsick	Out of work as temporarily sick	0.0042	0.0643	0	1
Hlimit1	Activities in last month not limited by health	0.7154	0.4513	0	1
Hlimit2	Rather limited	0.1751	0.3801	0	1
Hlimit3	Quite a lot	0.0669	0.2499	0	1
Hlimit4	A great deal	0.0426	0.2019	0	1
<i>Long-term health</i>					
H	self-reported health fair/poor	0.2528	0.4347	0	1

and the 1991 census data compare reasonably well. The total number of respondents is 5352, but due to missing information in some of the variables, especially the income variable, the sample size for the estimation of the models is reduced to 4814.²

The dependent count variable is the number of visits to or by a doctor (general practitioner) in the month before the interview. Variables that are included in the demand equation are sex, age, marital status, education, employment status, income, short-term health status, and the self-reported general health index. The binary health index, denoted H_i , is defined as

$$H_i = 1 \quad \text{if health is fair-poor}$$

$$H_i = 0 \quad \text{if health is excellent-good.}$$

Descriptions and summary statistics of the variables are given in Table I.

Every individual in the UK is covered by the National Health Service (NHS), which is paid for by payroll National Insurance contributions. A visit to a GP does not incur any additional cost,

² Sample selection bias could be modelled using the parametric techniques as proposed by Terza (1996) and discussed in Section 3.1.

Table II. Estimation results

Model Estimation Var	(A) Model 1 PL		(B) Model 1 GMM		(C) Model 2 PL two-stage	
	b	se	b	se	b	se
Const	-2.2752	0.3297	-2.5864	0.3298	-1.9693	0.3707
Male	-0.2748	0.0590	-0.2583	0.0620	-0.2953	0.0628
Single	-0.0104	0.1040	-0.0082	0.1064	-0.0388	0.1077
Age	0.0343	0.0124	0.0469	0.0125	0.0324	0.0129
Age ²	-0.0298	0.0114	-0.0418	0.0112	-0.0277	0.0118
Edu2	-0.2336	0.1518	-0.1984	0.1487	-0.1864	0.1564
Edu3	-0.2146	0.0853	-0.2206	0.0887	-0.1923	0.0883
Edu4	-0.2630	0.1200	-0.2974	0.1250	-0.2217	0.1228
Edu5	0.1477	0.1008	0.0759	0.1056	0.1718	0.1057
Unem	0.1416	0.1596	0.0939	0.1685	0.1559	0.1587
Inc2	0.1089	0.0644	0.0859	0.0646	0.1254	0.0649
Inc3	-0.1894	0.1080	-0.2534	0.1148	-0.1470	0.1128
Inc4	-0.5484	0.1920	-0.6924	0.2137	-0.5112	0.1937
Tempsick	0.5898	0.1423	0.6363	0.1401	0.5565	0.1501
Pregnant	0.9187	0.2025	0.8793	0.2202	1.0346	0.2351
Hlimit2	0.8191	0.0700	0.8917	0.0874	0.7629	0.0918
Hlimit3	1.1313	0.0895	1.2176	0.1232	1.0462	0.1281
Hlimit4	1.5032	0.1053	1.6230	0.1497	1.3621	0.1715
H	0.3832	0.0662	0.2448	0.2030		
H*					0.1112	0.0412
R^2	0.1910				0.1841	
	Test		DoF		p -value	
Hausman endogeneity test	0.5195		1		0.4711	
Overidentification test	47.61		34		0.0607	

Notes: The sample size is 4814. The dependent variable is 'Visits'. PL is the Poisson pseudo-likelihood estimator. GMM is iterated until convergence and is based on the instrument set $(F(z_i'\delta), z_i')$, with F the logistic CDF. The R^2 measure is based on deviance residuals (see Cameron and Windmeijer, 1996). The Hausman test for endogeneity is based on the coefficients of H_i . The test for overidentifying restrictions is the standard GMM χ^2 -test. H_i^* is predicted by $z_i'\delta$, with δ the logit estimator. The standard errors for the two-stage estimator are corrected for the estimation in stages (see Appendix).

and therefore no price variables are directly included in the model. Further, there is no information in the data set whether an individual has taken out a private insurance. In the UK, however, a person who has a private insurance remains covered by the NHS, and still pays the same contributions towards it. A private insurance tends to offer more choice at the elective specialist level, but less so at the GP level and it seems therefore likely that the non-availability of the information leads to only a minor misspecification. In 1991–2 approximately 14% of the population had a private insurance, with just under a half of this coverage provided by an employer (see Besley *et al.*, 1996).

In column (A) of Table II we first present Poisson pseudo-likelihood (PL) results for the model which includes the binary general health variable. The coefficient of H_i is 0.38 and is significant with a t -value of 5.78. As mentioned earlier, it is likely that the self-reported health index has a measurement error that is correlated with the number of visits to the doctor, as people who have

Table III. Instruments used in the health equation

Work status	Full time, part-time, permanently sick or disabled, retired, full time education, home
Social class	Seven classes
Accommodation	House, bungalow, other
Region	Eleven regions
Alcohol	Non-drinking, units taken last week (and squared) of beer, wine and spirits
Smoking	Ever smoked, smoke now
Long-term health	Long-standing illness, disability or infirmity
Other	Using oral contraceptive

recently visited a doctor may underreport their general health. In order to test whether the health index is endogenous, we specify Model 1 as³

$$y_i = \exp(\alpha H_i + x_i' \beta) + u_i \quad (9)$$

$$H_i^* = z_i' \delta + w_i \quad (10)$$

$$H_i = 1 \quad \text{if} \quad H_i^* \geq 0 \quad H_i = 0 \quad \text{otherwise}$$

The z_i are the instruments, which, apart from x_i , include variables that explain health, but are not likely to determine demand for doctors, other than via health. These variables are alcohol consumption and smoking behaviour, socio-economic status, regional variables, housing variables, work status and long-term health indicators, a total of 34 variables as summarized in Table III.

The equation for health, (10), is the reduced form for H_i^* . As argued in Section 3, the structural equation of H_i^* cannot be a function of the number of visits to the doctor for the system to be coherent. This does not seem an unreasonable specification for this particular application. H_i^* can be interpreted as long-term health, whereas the number of visits to the doctor is in the last month before interview. If there are any positive idiosyncratic shocks to *short*-term health this will be reflected by a higher than average number of visits to the doctor in that period. It may also lead to a perception of a person's own long-term health being worse than they would normally have reported. This means that the idiosyncratic shocks u and w in equations (9) and (10) are correlated, while the number of visits to the doctor does not have a direct effect on long-term health *per se*.

After logit estimation of the reduced form (10), a RESET-type test for misspecification and tests for heteroscedasticity did not indicate misspecification. Further, powers (of the few non-dummy variables) and cross-products that were considered most likely to contribute to the explanation of health were not jointly significant. Therefore, the linear reduced-form specification for H_i^* does not seem inappropriate.⁴

³ The health index was reported in categories 1–4, constituting the classes excellent, good, fair and poor. We initially set out by estimating an ordered logit and form the conditional mean of the index, or the conditional means of the separate categories as instruments. Using the four categories, however, resulted in very poor predictions, after which we decided to aggregate the excellent–good and fair–poor states.

⁴ The likelihood ratio test of the joint significance of the 34 instruments in the logit model has a value of 349.56 (p -value = 0.000). In comparison, this test statistic in the Poisson model with H_i included, is equal to 54.71 (p -value = 0.0136), with none of the instruments being individually significant (and $R^2 = 0.2010$). These results indicate that the instruments do explain health, but not visits to the doctor other than via health.

In column (B) of Table II, the results of the GMM estimator are presented, using as instruments $(F(z_i'\hat{\delta}), z_i')$, with F the logistic CDF. The value of the parameter of the health index is now equal to 0.24, a smaller value than previously, which is expected given the presumed positive correlation of H_i with the number of visits to the doctor. The difference, however, is not statistically significant as indicated by the Hausman test for endogeneity, comparing the estimated coefficient of H_i and its variance in column (B) with those in column (A). This result is due to the large estimated standard error of the coefficient in the GMM model. When the Hausman test is computed comparing all coefficients, the statistic is significant. This result, however, seems to arise from the fact that some standard errors are very close in both models. For example, when the constant is not considered in the test, the Hausman test statistic is insignificant. Further, some standard errors are smaller in the GMM model than in the PL model, giving an indefinite variance matrix for the difference in the parameters. Following an idea in Browning and Meghir (1991), we split the sample randomly into two equal subsamples and compared the estimation results for the PL model based on one subsample with the estimation results of the GMM model for the other subsample. The resulting Hausman test statistics were not significant.

Although the GMM estimator does not give conclusive evidence of the endogeneity of the self-reported health index, due to its relative imprecision, a comparison of the results of the additive specification to those of the multiplicative model (of which the full set of results are not presented here) gives some indication of endogeneity. First, the test for overidentifying restrictions rejects the null in the multiplicative model with a p -value of 0.00035, compared to a p -value of 0.0607 in the additive model. From the results of Section 2.1, this would indicate that z and u are uncorrelated, but that z and u/μ are correlated, which could occur when u and μ are correlated. Further, the pseudo-likelihood estimated coefficient of H_i in the multiplicative model was 0.4279, whereas the GMM estimated coefficient was *higher* and equal to 0.4975. This result seems implausible given the positive correlation between visits to the doctor and self-reported health. The two results together indicate that the z_i seem proper instruments for the additive model, whereas they are not for the multiplicative model.

Next, we proceed by estimating, in stages, the model that includes latent health H_i^* instead of H_i , which is perhaps the most appropriate way of modelling health. The estimation results are presented in column (C) of Table II. The first-stage parameter vector δ is estimated by the logit model, using the same reduced-form (10), and the second stage is estimated by Poisson pseudo-likelihood. The coefficient of H^* is equal to 0.11 and is significant. The results are quite similar to those in columns (A) and (B), but overall, the latent health specification adjusts the parameters more for the health effect. The results indicate that males visit the doctor less often than females. Marital status does not have an effect, and the age structure for demand is quadratic with a peak at around 57 years. This seems quite a low age, but this result is primarily driven by the conditioning on short-term health (Hlimit), which is positively correlated with age. Higher-educated people visit the doctor less frequently than lower-educated people, with the exception of the highest educated. Being unemployed does increase the demand, but this effect is not significant. The effect of income is slightly nonlinear, with the higher and lowest income groups having lower demand. The short-term health indicators are the most important in determining the demand for doctors in the last month before the interview. These results are broadly in line with, for example, those found for Australia by Cameron *et al.* (1988) and for Germany by Pohlmeier and Ulrich (1995).

5. CONCLUSIONS

This paper has examined the estimation of count data models with endogenous regressors, using the generalized method of moments (GMM) estimation technique. It has been shown that for model specifications with an additive or a multiplicative error term, the same set of instruments will not, in general, be orthogonal to both error types.

The GMM estimation technique was used to estimate a model for the explanation of the number of visits to the doctor by individuals. In the model specification, a self-reported health index was likely to be simultaneously determined. For nonlinear models it is often difficult to obtain the reduced form of the endogenous regressor as a simple function of the instruments. However, a simultaneous model of a count and a binary variable can only be logically consistent when the system is triangular. In the demand for doctors model, the binary health index was instrumented by the estimated probabilities of a reduced form logit model, which resulted in a smaller estimated coefficient when compared to the estimation by Poisson pseudo-likelihood. This difference was, however, not significant.

The model was finally specified in terms of continuous latent health, instead of the binary health index. This specification was estimated by a two-stage estimation procedure, using the same reduced-form logit estimates to form predictions of health.

APPENDIX: FORMULA FOR VARIANCE IN TWO-STAGE ESTIMATION PROCEDURE

Moment condition (8) can be written as

$$E[X^{*'}(y - \mu^*)] = 0$$

where

$$X^* = [X_2 \delta \ X_1] \quad \mu^* = \exp(\alpha X_2 \delta + X_1 \beta^*)$$

Let

$$\widehat{X}^* = [X_2 \hat{\delta} \ X_1] \quad \widehat{\mu}^* = \exp(\hat{\alpha} X_2 \hat{\delta} + X_1 \hat{\beta}^*)$$

where $\hat{\delta}$ is an initial consistent estimator for δ , and $\hat{\alpha}$ and $\hat{\beta}^*$ are the moment estimators, solving the sample moment condition

$$\widehat{X}^{*'}(y - \widehat{\mu}^*) = 0$$

Denote $\theta = (\alpha, \beta^{*'})'$, then a first-order Taylor series expansion, ignoring higher-order terms, results in

$$\widehat{X}^{*'}(y - \widehat{\mu}^*) = 0 = X^{*'}(y - \mu^*) - (X^{*'} M^* X^*)(\hat{\theta} - \theta) - (\alpha X^{*'} M^* X_2)(\hat{\delta} - \delta)$$

where $M^* = \text{diag}(\mu_i^*)$.

If $\hat{\delta}$ is the logit estimator, and if we let the variance of the count y_1 be general, then the asymptotic variance of $\hat{\theta}$ is given by

$$\begin{aligned} & (X^{*'} M^* X^*)^{-1} [X^{*'} \Omega^* X^* + \alpha^2 X^{*'} M^* X_2 (X_2' P X_2)^{-1} X_2' M^* X^* \\ & - \alpha X^{*'} M^* X_2 (X_2' P X_2)^{-1} X_2' W^* X^* - \alpha X^{*'} W^* X_2 (X_2' P X_2)^{-1} X_2' M^* X^*] (X^{*'} M^* X^*)^{-1} \end{aligned}$$

where

$$P = \text{diag}(\Lambda_i(1 - \Lambda_i)) \quad \Lambda_i = \frac{\exp(x'_{2i}\delta)}{1 + \exp(x'_{2i}\delta)}$$

$$\Omega = \text{diag}(E(y_{1i} - \mu_i^*)^2) \quad W^* = \text{diag}(E(y_{1i} - \mu_i^*)(y_{2i} - \Lambda_i))$$

and which is estimated by substituting the estimates $\hat{\alpha}$, $\hat{\beta}$ and $\hat{\delta}$ for α , β , and δ .

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